CHAPTER I

INTRODUCTION AND ITS BACKGROUND

Discuss the overview of the current state of technology and the current condition of the topic in the current local and international settings that are related to your chosen topic. This section provides the reader with the rationale of researchers' work and provides a sufficient background, relevance, and importance of the study.

This section serves as the introduction of the research paper, providing a brief overview of the project. A good introduction should contain a discussion of any or all of the following: (a) presentation of the problems (b) the existence of unsatisfactory condition, a felt problem that needs a solution to the gap of technology related to the study (c) show the rationale and justification of choosing the subject (d) historical or background of the problem.

The researchers must further explain the necessity for the software to be developed, and it should be in conformity to the gap in the technology that the researchers found out during the gap analysis. More importantly, this section should also present the algorithm/s to be used in the software and should be included in the discussion of the solutions.

TIP: Write here the introductory paragraph or opening statements about the main topic leading to your specific topic. This paragraph consists of 5 to 6 sentences. It starts with opening sentences about the main topic and detailed information on the succeeding sentences. A closing sentence is provided at the end of the paragraph.

The next paragraph is the discussion on the specific topic leading to your main problem. The main problem presents the reasons that motivate you to do research about this topic. Other specific problems that contributed to the main problem can be presented.

If the UN SDGs are a core motivation for your research, introduce them early in the introduction. You can incorporate a discussion of the relevant UN Sustainable Development Goals (SDGs) within the introduction section. This should include identifying which SDGs your research aligns with and explaining how your study contributes to addressing global challenges, promoting sustainable development, or advancing societal well-being.

Research Problem

A research problem is a statement about an area of concern, a condition to be improved, a difficulty to be eliminated, or a troubling question that exists in scholarly literature, in theory, or in practice that points to the need for meaningful understanding and deliberate investigation.

Also, it contains the general problem of the study that the proposed system will address. And it enumerates the specific problems that will be solved once the proposed system is completed. The next paragraph contains the causes and effects of the main problem presented. This paragraph should convince the reader that there is really an urgent need to solve or remedy the problem. The specific problems must consist of a minimum of 3 to 5 questions.

TIP: The next paragraph presents evidence that the problems really exist. Significant data can be presented, if any, and the sources of information must be

cited. Avoid presentation here of theoretical discussion, reserve them for the Review of Related Literature in Chapter 2.

Present here your intended solution to the problem and the corresponding strategy. Limit to one or two sentences only. Do not elaborate.

Research Objectives

It contains a general objective that will focus on what the study intends to achieve. It also enumerates the specific objectives that the researchers must undertake to meet the general objective. The formulated objectives should be answerable and aligned with the problems stated above.

Specifically, it aims to:

1.

Research Framework

Discuss the theoretical framework here followed by the conceptual framework. You may insert figures to graphically represent the frameworks. This section discusses the important theory/theories used by the researchers in the course of designing or developing the thesis. The theory/theories serve/s as the scientific anchor or as a support in conducting the study and should be derived from the Review of Literature. On this section, two frameworks are required to be used in developing the study. (1) the theoretical framework must be presented specifically in the discussion on how the algorithms will be utilized in software development. And (2) conceptual framework devised by researchers which was referenced from the theoretical framework should be presented and illustrated.

Theoretical Framework

TIP: When developing the Theoretical Framework (TF) for your research proposal, you should focus on identifying and discussing the existing theories, models, or principles that underpin your study. This involves reviewing relevant scholarly literature to highlight established concepts that provide a foundation for understanding your research problem. TF serves to situate your study within the broader academic context by explaining how these theories relate to your research questions and hypotheses. It should be grounded in well-recognized theoretical perspectives and demonstrate how they inform your approach, guiding the interpretation of your findings.

EXAMPLE: if your research involves cybersecurity, you might reference theories related to cryptography or threat detection models. If it pertains to software engineering, you could discuss development methodologies or quality assurance frameworks.

Conceptual Framework

A conceptual framework represents the researcher's synthesis of literature on how to explain a phenomenon. It maps out the actions required in the course of the study given his previous knowledge of other researchers' point of view and his observations on the subject of research. Build your conceptual framework using your mix of the variables from the scientific articles you have read. Your problem statement serves as a reference for constructing the conceptual framework. In effect, your study will attempt to answer a question that other researchers have not explained yet. Your research should address a knowledge gap.

TIP: the Conceptual Framework (CF) is more focused on your specific research context and the relationships among variables you plan to examine. It is a visual or narrative representation of your research design, outlining the key concepts, variables, and their presumed interactions based on your understanding and review of the literature. CF helps clarify the scope and focus of your study by illustrating how you conceptualize the problem and the expected relationships or effects within your specific setting. Unlike TF, which relies on established theories, CF is more tailored to your research, often constructed by

synthesizing relevant concepts and defining operational relationships that will be empirically tested.

EXAMPLE: If you are studying the impact of a new machine learning algorithm on data classification accuracy, your CF would illustrate the variables involved—such as algorithm type, dataset features, and accuracy metrics—and how they relate based on your hypothesis.

Scope and Limitations of the Research

The scope and limitation include the brief statement of the general purpose of the study and the subject matter and topic studies discussed. Also, the population or universe from which the respondents are selected. This must be large enough to make the generalization significant. The scope identifies the coverage of the study in terms of subject, facilities, objectives, area, time duration of the issues to which the research is focused.

In the limitation of the study, this section must include the weaknesses of the study that are beyond the control of the researchers. The limitation of the study defines the constraints or weaknesses which are not within the control of the writer, therefore they are not expected to be covered by the study.

Significance of the Research

This section must explain the coverage of why the research focuses on this area, the expected contributions and who will benefit from the research.

If your study aims to contribute toward achieving specific SDGs, you can elaborate on this in the significance section. Here, you can discuss how your

research benefits broader societal goals and aligns with global development agendas. Highlight the potential impact of your findings on addressing global challenges, contributing to sustainable development, or improving societal well-being.

The beneficiaries should be stated from a macro to micro perspective. You may follow the format below:

TIP: This study will benefit the _____.

The benefits that can be derived from the study include the following:

Example (**Local Government Unit)**. Include a description of the significant contributions the study is expected to make to the beneficiaries.

Example (**Residents)**. Include a description of the significant contributions the study is expected to make to the beneficiaries.